Potential binary supermassive black holes in Markarian 231 as a scientific case for the spectropolarimeter POLLUX

Julie Biedermann [1] Frédéric Marin [1]

[1]Astronomical Observatory of Strasbourg, Strasbourg University, CNRS

Active Galactic Nuclei

Supermassive black holes (SMBHs), with masses ranging from 10^6 to 10^9 $M_\odot$, are thought to reside at the center of nearly all galaxies. In the context of galaxy mergers, the SMBHs from each progenitor are expected to form a binary system. Given the frequency of such mergers, **binary SMBHs** should be common in active galactic nuclei (AGNs). However, their detection remains challenging and relies mainly on indirect signatures.

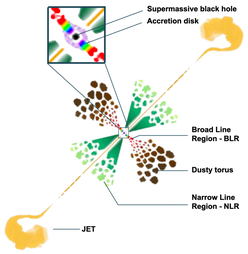


Figure 1. Schematic structure of an AGN. [1].

Quasars are the most luminous subclass of AGNs, often outshine their host galaxies. Their central engine consists of an **SMBH** surrounded by an **accretion disk** emitting strong UV-optical radiation. In the equatorial plane, there is a **dusty torus** enveloping this structure, which can obscure the nucleus depending on the viewing angle. Relativistic **jets** may emerge along the polar axis, sometimes extending over megaparsec scales. The **broad-line region** (BLR), located just above the disk, contains fast-moving dense gas clouds (1000–250 000 km.s^{-1}), while the more extended **narrow-line region** (NLR) is composed of lower-density ionized gas with velocities below 1000 km.s^{-1}. This geometry is schematically illustrated in Figure 1.

Scientific case : Markarian 231

Mrk 231 (z = 0.0422) is a quasar located in an ultraluminous infrared galaxies (ULIRGs) in the local universe. Its disturbed morphology and tidal features suggest an advanced-stage galactic merger, making it a strong candidate for hosting a supermassive black hole binary.

We compiled archival spectropolarimetric data from the UV to mid-IR, displaying in Figure 2. Its UV-optical spectrum exhibits unique characteristics :

- A sharp **drop in UV flux** around 2500 Å, followed by a slow rise at shorter wavelengths. This deviates from typical quasar spectra and may reflect a circumbinary disk with an opening gap created by a secondary SMBH [5].
- Strongly **wavelength-dependent polarization degree** : up to ∼20% in the UV, dropping below 0.5% in the IR (see Fig. 2, third panel).
- A smooth **rotation of the polarization position angle** across the same waveband region where we observe the break in the flux, suggesting complex scattering geometry and possibly two emission sources.

Our goal is to test the SMBH binary scenario by modeling both total and polarized fluxes using a simple analytical approach, and comparing the results with the observed spectropolarimetric signatures.

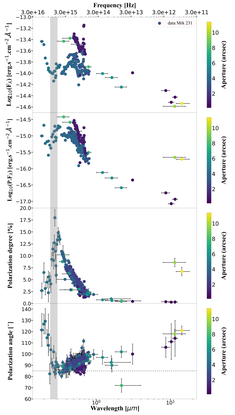


Figure 2. Spectropoalrimetric data of Mrk 231. The aperture of the observations is color-coded (in arcseconds).

Method – SED of a Binary SMBH System

Objective: Reproduce the total Spectral Energy Distribution (SED) of Mrk 231 using a binary SMBH model proposed by Yan et al. [5], where a massive primary SMBH dominates in the optical, while a less massive secondary SMBH contributes mainly in the far-UV.

Primary SMBH: Modeled using a Shakura-Sunyaev thin, optically optically thick and geometrically thin accretion disk, emitting as a multi-temperature blackbody.

- *Inner radius:* Set by the Hill radius, which depends on the binary semi-major axis $a_{\rm BBH}$ and mass ratio q.
- *Outer radius:* Defined by the Toomre stability criterion, marking where the disk becomes gravitationally unstable and depends on the accretion rate of the system.

Secondary SMBH: Located within the primary disk, it opens an inner gap and accretes via a mini-disk.

- *Inner radius:* Set to the ISCO of a non-rotating black hole.
- *Outer radius:* Fraction of the Roche lobe radius (function of q and $a_{\rm BBH}$).

Additional components: To reproduce the full observed SED of Mrk 231, we added emission components corresponding to the main quasar structures, selecting templates that match the observed age and luminosity of Mrk 231.

- *Host galaxy:* Modeled using a 13 Gyr elliptical template [3].
- *Dusty torus:* Infrared emission added using torus models from [4].

The full model (binary + torus + host) is shown in Fig. 3 (top left panel).

Results

Fig. 3 shows the best-fit models for both total flux and polarizations of Mrk 231. The model predicts a second bump in the UV in the polarized flux, which corresponds to the emission from the smaller secondary SMBH in our binary scenario.

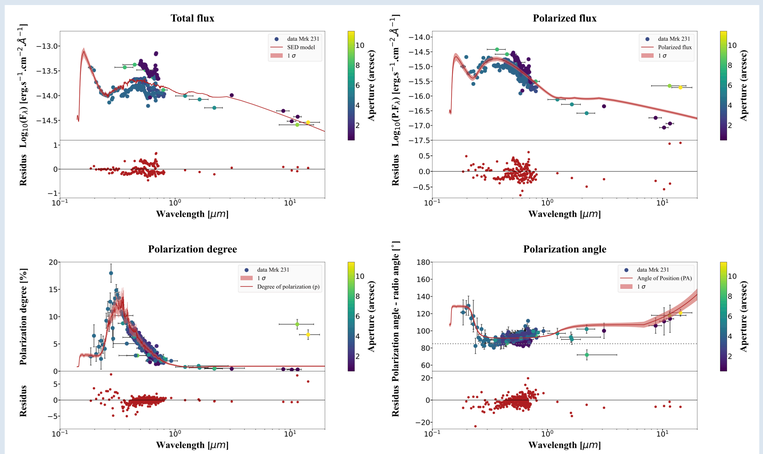


Figure 3. Results of the model fitting for Mrk 231.

Method - Polarimetric Modeling

Objective: Modeled the polarimetric response of our SED without modifying the geometrical configuration of the system.

Polarization modeling: Each AGN component (primary and secondary SMBHs, dusty torus and host galaxy) was assigned a wavelength-independent linear polarization, defined by Stokes parameters (I, Q, U, V).

- Total intensity I_i was fixed from the best-fit flux $F_{\lambda,i}$ of the SED model.
- Linear polarization: $Q_i = q_i \times I_i$ and $U_i = u_i \times I_i$
- No circular polarization: $V_i = 0$ (no data available for Mrk 231)

Fitting procedure: We optimized the polarization fractions (q_i, $u_i \in [-1, 1]$) to fit the observed total Stokes Q and U from Mrk 231.

Output: From the summed Stokes parameters, we computed:

- Polarized flux (Fig 3, top-right)
- Degree of polarization (Fig 3, bottom-left)
- Polarization angle (Fig 3, bottom-right)

Conclusion

Our results support the **binary SMBH scenario for Mrk 231** by successfully reproducing three independent observables: total flux, polarization degree, and polarization angle.

The model predicts a second bump in the far-UV polarized flux corresponding to the contribution of the smaller secondary SMBH. However, the lack of far-UV spectropolarimetric data prevents direct confirmation.

This study highlights the promising potential of future space-based UV spectropolarimetry to probe the inner structures of quasars and test the binary SMBH scenario. Especially, the upcoming instrument **POLLUX**, a high-resolution UV spectropolarimeter proposed by an European consortium for the HWO, could provide key constraints [2]. Covering the 100 nm to 1.5 μm range with unprecedented spectral resolution ; R > 60,000, POLLUX will allow detailed measurements of polarized light in the far-UV, where binary SMBH signatures are most prominent. Based on our results, the following observational requirements are identify to confirm the binary SMBH scenario in Mrk 231 and similar objects:

→ Hight spectral resolution (R>20 000) in the Ly region to monitor potential variability.

→ Far-UV polarimetric flux (down to 1000 Å) to confirm or reject the presence of the second bump.

→ Use of UV spectropolarimeters as a powerful tool for detecting binary SMBHs in AGNs.

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